

Maloney's Geomethry

An Axiomatic Formalization of Synthetic Geometry

$$\mathfrak{M} := \langle \mathbf{P}, \mathcal{B}, \cong, \mathcal{L}, \mathcal{A} \rangle$$

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Guide to Notation

Objects and Sets

$A, B, C, \dots, X, Y, Z$

Capital italic: **points**

P The set of all points

$\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \dots \mathfrak{X}, \mathfrak{Y}, \mathfrak{Z}$ Capital x: geometric figures

$\overline{AB}$ **Line Segment**

$\overleftrightarrow{AB}$ **Line**

$\overrightarrow{AB}$ **Ray**

$\triangle ABC$ **Triangle**

$\angle ABC$ **Angle**

$\overline{AB} \parallel \overline{CD}$ **Parallel**

$\overline{AB} \perp \overline{CD}$ **Perpendicular**

0.1 Primitive Relations

$\mathcal{B}(A, B, C)$ Betweenness

$A = B$ Equivalence

$\overline{AB}, \overline{CD}$ Congruence

0.2 idk

$\mathcal{L}$ The **Language**

$\mathcal{A}$ The **Axiom Set**

Part I

Philosophical Foundations

Chapter 1

The Ontology of Geometry

1.1 Existence: The Reality of Mathematical Objects

Is math real? Discussing Platonism vs. Nominalism.

1.2 Origin: Discovery and Invention

1.3 Modality: Necessity, Possibility, and Contingency

1.4 Identity: The Nature of Objects and Relations

Chapter 2

Epistemology of Mathematics

2.1 The Nature of Truth

2.2 Axioms, Theorems, Structure

2.3 Definitions

2.4 Proof, Justification, and Rigor

Part II

Logical Foundations

Chapter 3

First-Order Logic

3.1 Logical Syntax

3.2 Logical Connectives

3.3 Variables and Quantifiers

3.4 Rules of Inference and Deduction

Chapter 4

The Signature of Geometry

4.1 Formal Signatures

Specification of primitive symbols and arities defining a formal system.

Definition 1 (Signature of Geometry). *A signature $\mathcal{L}_{geo}$ consists of:*

- *a ternary relation symbol $\mathcal{B}$ (betweenness),*
- *a quaternary relation symbol $\cong$ (congruence).*

4.2 Geometric Language

Formal language generated from geometric primitives and relation symbols.

4.3 Typed Signature Extension

Signatures enriched with dependent types to encode geometric constraints directly in syntax.

4.4 The Geometric Language and Well-Formed Formulas

Chapter 5

The Axiomatic System

5.1 Primitve Objects

5.2 Primitve Relations

5.3 Axioms

5.4 Derived Notions

5.5 Derived Proofs

5.6 Models and Structures

Part III

Set-Theoretic Geometry

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Chapter 6

Primitive Notions

For our axiomatic system, we adopt three primitive notions, consisting of one object and two relations, which together form a single structure. The first of these is the point, denoted by upper case latin letters, $A, B, C, \dots$. Each point is a member of the point-set $\mathbf{P}$, which is defined as $\mathbf{P} := \{P \mid P \text{ is a point}\}$. This set is equipped with a ternary relation $\mathcal{B}$ (betweenness) and an quaternary relation $\cong$ (congruence). Together, the domain and these two relations constitute the plane $\mathcal{P}$:

$$\mathcal{P} := \langle \mathbf{P}, \mathcal{B}, \cong \rangle.$$

6.1 Plane

The plane is the structure consisting of the domain $\mathbf{P}$ together with the relations $\mathcal{B}$ and $\cong$.

Definition 2 (Plane). *A plane is a structure*

$$\mathcal{P} := \langle \mathbf{P}, \mathcal{B}, \cong \rangle.$$

Where $\mathbf{P}$ is the domain.

6.2 Points

The elements of the domain $\mathbf{P}$ are termed *points* and denoted using capital italics such as $A, B, C, \dots$. At this foundational level, a point is defined not via decomposition into simpler parts, but solely by its membership in this set. characterization Intuitively, a point can be understood as a unique position on the plane. This corresponds to the Euclidean characterization of a point as "that which has no part". A point can be informally regarded as having no magnitude, length, width, or depth, and as being zero-dimensional. However, within the formal system, this physical characterization is set aside and a point is treated strictly as a primitive object with no further definition, thus, the characterization of these objects carries no weight in the formal deduction.

6.3 Betweenness

Betweenness is a ternary relation on the set of all points, denoted by $\mathcal{B} \subseteq \mathbf{P}^3$. Given points A, B , and C , we write $\mathcal{B}(A, B, C)$ and read it as " B lies between A and C ".

Unlike a set $\{A, B, C\}$, where the order of elements is immaterial, betweenness is defined over the cartesian product $\mathbf{P} \times \mathbf{P} \times \mathbf{P}$. Thus it depends on the order of the arguments. Formally,

$$\mathcal{B} \subseteq \mathbf{P}^3 := \{(A, B, C) \mid A, B, C \in \mathbf{P}\}.$$

The expression $\mathcal{B}(A, B, C)$ is simply the assertion that $(A, B, C) \in \mathcal{B}$.

In Euclidean geometry a point B is between A and C precisely when the three points lie on a line and B is met while traveling from A to C along that line. Our axioms capture this idea without invoking lines or order directly.

We now x five axioms governing the behavior of $\mathcal{B}$.

6.3.1 Axioms of Betweenness

Axiom 1 (Symmetry of Betweenness).

$$\forall A, B, C \in \mathbf{P}, \mathcal{B}(A, B, C) \implies \mathcal{B}(C, B, A).$$

Betweenness is symmetric with respect to the outer points: if B lies between A and C , then it also lies between C and A . This reflects the fact that the segment AC is the same as the segment CA . Left-to-right implication is used over a biconditional due to redundancy, as the right-to-left implication directly follows from this form of the axiom.

Axiom 2 (Order). $\forall A, B, X \in \mathbf{P}, \mathcal{B}(A, B, X) \implies (\neg \mathcal{B}(B, A, X) \wedge \neg \mathcal{B}(A, X, B)).$

If B is between A and X , then A cannot be between B and X , nor can X be between A and B . In other words, a point can serve as a “middle” in only one way: the triple (A, B, X) determines a unique order along the line. Together with the Identity axiom, this implies that $\mathcal{B}(A, A, A)$ is impossible (otherwise, taking $A = B = X$, we would have both $\mathcal{B}(A, A, A)$ and its negation). Thus the relation is *non-reflexive* in the strict sense: no point lies between itself and itself.

Axiom 3 (Identity of Betweenness).

$$\forall A, B \in \mathbf{P}, \mathcal{B}(A, B, A) \implies A = B.$$

This axiom says that if a point appears as the “middle” of a triple that begins and ends with the same point, then the two outer points must coincide. In particular, it forbids the possibility that a distinct point B lies between A and A ; such a configuration would contradict the intuition that betweenness requires a genuine interval. Notice that the converse, $A = B \implies \mathcal{B}(A, A, A)$, is *not* assumed; indeed the next axiom will exclude it.

Axiom 4 (Density). $\forall A, C \in \mathbf{P}, A \neq C \implies \exists B \in \mathbf{P}: \mathcal{B}(A, B, C).$

Whenever two points are distinct, there exists a point strictly between them. This captures the idea that segments are dense—between any two points one can always find a third.

Axiom 5 (Extensibility).

$$\forall A, B \in \mathbf{P}, A \neq B \implies \exists C \in \mathbf{P}: \mathcal{B}(A, B, C).$$

Given two distinct points, there is always a point beyond the second, so that the second lies between the first and the new point. This guarantees that lines extend indefinitely.

Axiom 6 (Transitivity of Betweenness). $\mathcal{B}(A, B, C) \wedge \mathcal{B}(B, C, D) \wedge B \neq C \implies \mathcal{B}(A, B, D).$

If B lies between A and C , and C lies between B and D (with $B \neq C$), then B lies between A and D . This axiom allows us to chain betweenness relations along a line.

These six axioms are mutually independent: no one can be derived from the others (as can be shown by standard model-theoretic arguments). Together with the congruence axioms to be introduced later, they will form the basis of our geometric theory.

Axiom 7. $\exists A, B \in \mathbf{P}, A \neq B.$

6.3.2 Immediate Consequences

Theorem 1. $\mathcal{B}(A, B, C) \iff \mathcal{B}(C, B, A)$ for all $A, B, C \in \mathbf{P}$.

Proof. The forward direction is the axiom; the reverse follows by applying the axiom with A and C exchanged. $\square$

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Given two distinct points, there is always a point beyond the second, so that the second lies between the first and the new point. This guarantees that lines extend indefinitely.

These five axioms are mutually independent: no one can be derived from the others (as can be shown by standard model-theoretic arguments). Together with the congruence axioms to be introduced later, they will form the basis of our geometric theory.

Theorem 2 (Irreflexivity of Betweenness). *No point lies between it and itself:*

$$\forall A \in \mathbf{P}, \neg \mathcal{B}(A, A, A).$$

Proof. Assume $\mathcal{B}(A, A, A)$. By the order axiom, this implies $\neg \mathcal{B}(A, A, A)$ and $\neg \mathcal{B}(A, A, A)$, which contradicts our assumption. Hence, $\neg \mathcal{B}(A, A, A)$. $\square$

Theorem 3 (Distinct Endpoints). *Betweenness forces distinct outer points:*

$$\mathcal{B}(A, B, C) \implies A \neq C.$$

Proof. Suppose $\mathcal{B}(A, B, C)$ and $A = C$. Then, we have $\mathcal{B}(A, B, A)$, which by the identity of betweenness implies $A = B$. Substituting back gives $\mathcal{B}(A, A, A)$, which contradicts the irreflexivity of betweenness. Therefore $A \neq C$. $\square$

Theorem 4 (Strict Betweenness). *If $\mathcal{B}(A, B, C)$, then $A \neq B$ and $B \neq C$.*

Proof. Suppose $\mathcal{B}(A, B, C)$ and $A = B$. Then, $\mathcal{B}(A, A, C)$, which implies $\neg \mathcal{B}(A, A, C)$ by the order of betweenness. However, this is a contradiction. Thus, $A \neq B$. Similarly, if we assume $B = C$, then we have $\mathcal{B}(A, B, B) \implies \neg \mathcal{B}(A, B, B)$, leading to a contradiction. Thus, $B \neq C$. $\square$

Theorem 5 (Contrapositive Ordering). *For all $A, B, X \in \mathbf{P}$,*

$$\mathcal{B}(B, A, X) \implies \neg \mathcal{B}(A, B, X) \quad \text{and} \quad \mathcal{B}(A, X, B) \implies \neg \mathcal{B}(A, B, X).$$

Proof. The Order axiom gives the two implications

$$\mathcal{B}(A, B, X) \implies \neg \mathcal{B}(B, A, X) \quad \text{and} \quad \mathcal{B}(A, B, X) \implies \neg \mathcal{B}(A, X, B).$$

Taking the logical contrapositive of each implication (which is logically equivalent to the original), yields precisely the two stated results. $\square$

Theorem 6 (Symmetry Extended). *Given $\mathcal{B}(A, B, C)$ none of the following hold:*

$$\mathcal{B}(B, A, C), \mathcal{B}(A, C, B), \mathcal{B}(B, C, A), \mathcal{B}(C, A, B).$$

6.3.3 Infinite Plane

Theorem 7 (Outer Transitivity Variant). *For all $X, Y, Z, W \in \mathbf{P}$,*

$$\mathcal{B}(X, Y, Z) \wedge \mathcal{B}(X, Z, W) \wedge Y \neq Z \implies \mathcal{B}(X, Y, W).$$

Proof. Assume $\mathcal{B}(X, Y, Z)$, $\mathcal{B}(X, Z, W)$, and $Y \neq Z$. From $\mathcal{B}(X, Y, Z)$ and $Y \neq Z$, Symmetry gives $\mathcal{B}(Z, Y, X)$. From $\mathcal{B}(X, Z, W)$, Symmetry gives $\mathcal{B}(W, Z, X)$. Now apply the Transitivity axiom (Axiom ??) with $(A, B, C, D) = (W, Z, Y, X)$:

$$\mathcal{B}(W, Z, X) \wedge \mathcal{B}(Z, Y, X) \wedge Z \neq Y \implies \mathcal{B}(W, Y, X).$$

The condition $Z \neq Y$ holds by assumption. Applying Symmetry to $\mathcal{B}(W, Y, X)$ yields $\mathcal{B}(X, Y, W)$, as required. $\square$

Theorem 8 (Infinite outward extension). *Let $A, B \in \mathbf{P}$ with $A \neq B$. Then there exist infinitely many points $C_1, C_2, C_3, \dots$ such that*

$$\mathcal{B}(A, B, C_1), \mathcal{B}(A, C_1, C_2), \mathcal{B}(A, C_2, C_3), \dots$$

Proof. $\square$

These theorems complete our baseline analysis of how the betweenness relation $\mathcal{B}$ operates locally on individual triples of points. However, proving that the plane $\mathcal{P}$ is structurally infinite—both in its internal density and its outward extension—requires the formal definitions of segments and lines. We defer these proofs to Chapter ??, where the infinite nature of the geometric line can be properly established.

6.4 Congruence

Congruence is a quaternary relation on the set of all points denoted by $\cong \subseteq \mathbf{P}^4$. Rather than mapping individual points to one another, this relation establishes a geometric equivalence between two pairs of points. We write $\cong (A, B, C, D)$, to assert that the segment from point A to point B is congruent to the segment from point C to point D . Formally:

$$\cong (A, B, C, D) \iff (A, B, C, D) \in \cong.$$

To reflect this conceptually, we bundle the point pairs (A, B) and (C, D) into the typographical units AB and CD and introduce the infix symbol $\cong$:

$$AB \cong CD \stackrel{\text{def}}{\iff} \cong (A, B, C, D)$$

If the relation holds for the pairs (A, B) and (C, D) , we write $(A, B) \cong (C, D)$, which is interpreted as the "distance" from A to B being equal to the distance from C to D .

We now impose three axioms governing the behavior of $\cong$.

Axiom 8 (Reversal of Congruence).

$$\forall A, B \in \mathbf{P}, AB \cong BA.$$

Axiom 9 (Reflexivity of Congruence).

$$\forall A, B \in \mathbf{P}, AB \cong AB.$$

Axiom 10 (Transitivity of Congruence).

$$\forall A, B, C, D, E, F \in \mathbf{P}, (AB \cong CD \wedge AB \cong EF) \implies CD \cong EF.$$

Theorem 9 (Symmetry of Congruence). *If $AB \cong CD$, then $CD \cong AB$.*

Proof. We are given $AB \cong CD$. From the reflexivity of congruence axiom we also have $AB \cong AB$. Applying these two congruences to the transitivity axiom gives the antecedent

$$AB \cong CD \wedge AB \cong AB,$$

which implies $CD \cong AB$. □

Theorem 10 (Left-Reversal of Congruence). *If $AB \cong CD$, then $BA \cong CD$.*

Proof. We are given $AB \cong CD$. From the reversal of congruence, we also have $AB \cong BA$. Applying transitivity to the conjunction

$$AB \cong BA \wedge AB \cong CD$$

yields $BA \cong CD$. □

Theorem 11 (Right-Reversal of Congruence). *If $AB \cong CD$, then $AB \cong DC$.*

Proof. We are given $AB \cong CD$. From the reversal of congruence, we have $CD \cong DC$, and by the symmetry of congruence we have $CD \cong AB$. Applying transitivity to

$$CD \cong AB \wedge CD \cong DC,$$

yields $AB \cong DC$. □

Theorem 12 (Full-Reversal of Congruence). *If $AB \cong CD$, then $BA \cong DC$.*

Proof. From $AB \cong CD$, left-reversal gives $BA \cong CD$. Applying right-reversal to $BA \cong CD$ yields $BA \cong DC$. □

Theorem 13 (Full-Transitivity of Congruence). *If $AB \cong CD$ and $CD \cong EF$, then $AB \cong EF$.*

Proof. We first apply symmetry to the first premise, giving $CD \cong AB$. Then, we apply this result along with the second premise to the transitivity axiom, giving the antecedent

$$CD \cong AB \wedge CD \cong EF.$$

This implies $AB \cong EF$, thus proving the result. □

With these results, we have demonstrated that $\cong$ is an equivalence relation on ordered pairs of points, as it is reflexive (by axiom) and symmetric (by theorem), and fully transitive (by theorem).

Because $\cong$ is an equivalence relation, it partitions the set of all ordered point pairs into disjoint equivalence classes. Each class consists of all segments that are congruent to one another, which in geometric terms corresponds to the set of all segments having the same length. This formally captures the intuitive idea that congruence of segments is equality of distance.

Chapter 7

Linear Notions

Theorem 14 (Infinite interior points). *Let $A, C \in \mathbf{P}$ with $A \neq C$. Then there exist infinitely many points strictly between A and C .*

Proof. wasd □

Theorem 15 (Infinite outward extension). *Let $A, B \in \mathbf{P}$ with $A \neq B$. Then there exist infinitely many points $C_1, C_2, C_3, \dots$ such that*

$$\mathcal{B}(A, B, C_1), \quad \mathcal{B}(A, C_1, C_2), \quad \mathcal{B}(A, C_2, C_3), \quad \dots$$

Proof. wasd □

Together, these two lemmas show that from any segment AB ($A \neq B$) we can generate infinitely many points in both directions along the line determined by A and B . Thus the plane $\mathcal{P}$ is necessarily infinite.

7.1 Defintion of Linear Notions

A segment is a subset of $\mathbf{P}$ defined as the union of two discint points and all points between them. Symbolically, that is

Definition 3 (Segment).

$$\forall A, B \in \mathbf{P}, A \neq B \implies AB := \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}.$$

A ray pointing from point A to a discint point B is a subset of $\mathbf{P}$ denoted as $\overrightarrow{AB}$. It is defined as the union of A, B , all points between A and B , and all points such that B is between A and that point, that is

Definition 4 (Ray).

$$\forall A, B \in \mathbf{P}, A \neq B \implies \overrightarrow{AB} := \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

The line $\overleftrightarrow{AB}$ is the subset of $\mathbf{P}$ consisting of the points A, B , all points between A and B , all points where A is between it and B , and all points where B is between it and A . That is,

Definition 5 (Line).

$$\forall A, B \in \mathbf{P}, A \neq B \implies \overleftrightarrow{AB} := \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

7.2 Properties of Linear Notions

Theorem 16 (Distinctness of Betweenness). *If $\mathcal{B}(A, B, X)$ holds, then A , B , and X are three distinct points:*

$$\forall A, B, X \in \mathbf{P}, \mathcal{B}(A, B, X) \implies (A \neq B \wedge B \neq X \wedge A \neq X).$$

Proof. We proceed by contradiction. Suppose $\mathcal{B}(A, B, X)$ holds.

First, assume $A = X$. Substituting A for X in our hypothesis, we obtain $\mathcal{B}(A, B, A)$, which implies $A = B$. If $A = B$, then the relation $\mathcal{B}(A, B, X)$ is equivalent to $\mathcal{B}(B, A, X)$. However, by the Order of Betweenness, $\mathcal{B}(A, B, X) \implies \neg \mathcal{B}(B, A, X)$. Here, we reach a contradiction and conclude our hypothesis must be false and thus $A \neq X$.

Next, we test the assumption $A = B$. Under this assumption, we substitute our hypothesis to obtain $\mathcal{B}(B, A, X)$. Similarly, the Order of Betweenness shows these cannot be simultaneously true, and thus our assumption must be false, meaning $A \neq B$.

Finally, by the Transitivity of Equality, $B \neq X$. Thus proving the result. $\square$

Theorem 17 (Segment Symmetry). *For any two points A, B , the segment AB is equal to the segment BA .*

$$\forall A, B \in \mathbf{P}, AB = BA.$$

Proof. By the definition of a line segment, we have:

$$AB = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}$$

and

$$BA = \{B, A\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A)\}.$$

First, by the commutative property of sets, the endpoints are identical: $\{A, B\} = \{B, A\}$.

Second, we examine the sets of interior points. By the **Symmetry of Betweenness**, the relation $\mathcal{B}(A, X, B)$ holds if and only if $\mathcal{B}(B, X, A)$ holds. Formally, this implies:

$$\{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\} = \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A)\}.$$

Since AB and BA are defined as the union of identical sets, they contain the same points. By the principle of extensionality in set theory, $AB = BA$. $\square$

Theorem 18 (Line Symmetry). *For any two distinct points A and B , the line $\overleftrightarrow{AB}$ is equal to the segment $\overleftrightarrow{BA}$:*

$$\forall A, B \in \mathbf{P}, \overleftrightarrow{AB} = \overleftrightarrow{BA}.$$

Proof. By the definition of a line, we have:

$$\overleftrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}$$

and

$$\overleftrightarrow{BA} = \{B, A\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, B, A) \vee \mathcal{B}(B, X, A) \vee \mathcal{B}(B, A, X)\}.$$

First, by the commutative property of sets, the endpoints are identical: $\{A, B\} = \{B, A\}$.

Second, we demonstrate that the three cases of betweenness in $\overleftrightarrow{AB}$ are logically equivalent to those of $\overleftrightarrow{BA}$ using the **Symmetry of Betweenness**:

- $\{X \in \mathbf{P} \mid \mathcal{B}(X, A, B)\} = \{X \in \mathbf{P} \mid \mathcal{B}(B, A, X)\}$
- $\{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\} = \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A)\}$
- $\{X \in \mathbf{P} \mid \mathcal{B}(A, B, X)\} = \{X \in \mathbf{P} \mid \mathcal{B}(X, B, A)\}$

Since each condition for membership in $\overleftrightarrow{AB}$ is logically equivalent to a condition for membership in $\overleftrightarrow{BA}$, the sets contain the same points. Thus, by the principle of extensionality, $\overleftrightarrow{AB} = \overleftrightarrow{BA}$. $\square$

Theorem 19 (Ray Asymmetry). *For any two distinct points $A, B \in \mathbf{P}$, the ray $\overrightarrow{AB}$ is not equal to the ray $\overrightarrow{BA}$:*

$$\forall A, B \in \mathbf{P}, A \neq B \implies \overrightarrow{AB} \neq \overrightarrow{BA}.$$

Proof. We proceed by contradiction. Assume that the rays for some distinct points $A, B \in \mathbf{P}$, $\overrightarrow{AB} = \overrightarrow{BA}$.

By the definition of a ray, we have:

$$\overrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\},$$

and

$$\overrightarrow{BA} = \{B, A\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A) \vee \mathcal{B}(B, A, X)\}.$$

By the **Outer Point Axiom**, there must exist a point Y such that $\mathcal{B}(A, B, Y)$. By the definition of $\overrightarrow{AB}$, it follows that $Y \in \overrightarrow{AB}$. Under our initial assumption that the rays, Y must be an element of $\overrightarrow{BA}$.

By the Distinctness of Betweenness, the relation $\mathcal{B}(A, B, Y)$ implies that $A \neq Y$ and $B \neq Y$. Therefore, $Y \notin \{A, B\}$.

Since $Y \in \overrightarrow{BA}$ but is not an endpoint, it must satisfy $\mathcal{B}(B, Y, A)$ or $\mathcal{B}(B, A, Y)$.

However, because we know that $\mathcal{B}(A, B, Y)$ is true, by Order of Betweenness we have $\mathcal{B}(A, B, Y) \implies \neg \mathcal{B}(B, A, Y)$, and $\mathcal{B}(A, B, Y) \implies \neg \mathcal{B}(A, Y, B)$, which by the Symmetry of Betweenness is logically equivalent to $\mathcal{B}(A, B, Y) \implies \neg \mathcal{B}(B, Y, A)$.

Thus, both conditions for Y to be an element of $\overrightarrow{BA}$ are false. Therefore, the rays $\overrightarrow{AB}$ and $\overrightarrow{BA}$ are not equal, thus proving the desired result. $\square$

Theorem 20 (Union of Opposite Rays). *For any two points $A, B \in \mathbf{P}$,*

$$\overleftarrow{AB} \cup \overrightarrow{AB} = \overleftrightarrow{AB}.$$

Proof. By the definition of a ray, we have:

$$\overleftarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B)\},$$

and

$$\overrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

The union of $\overleftarrow{AB}$ and $\overrightarrow{AB}$ therefore consists of the set, $\{B, A\}$, and all points X such that $\mathcal{B}(X, A, B)$ or $\mathcal{B}(A, X, B)$, or $\mathcal{B}(A, B, X)$.

Note that this is the same as the line $\overleftrightarrow{AB}$, as it is defined as the set

$$\overleftrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

This set is equal to the union described above. Thus, the union of the rays $\overleftarrow{AB}$ and $\overrightarrow{AB}$ is equal to the line $\overleftrightarrow{AB}$. $\square$

Theorem 21 (Intersection of Opposite Rays). *For any two points $A, B \in \mathbf{P}$,*

$$\overleftarrow{AB} \cap \overrightarrow{AB} = AB.$$

Proof. By the definition of a ray, we have:

$$\overleftarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B)\},$$

and

$$\overrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

The intersection of $\overleftarrow{AB}$ and $\overrightarrow{AB}$ therefore consists of the set, $\{B, A\}$ and all points X between A and B , that is $\mathcal{B}(A, X, B)$.

This intersection consists of the same points contained in the segment AB , as

$$AB = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}.$$

Thus, the union of the rays $\overleftarrow{AB}$ and $\overrightarrow{AB}$ is equal to the segment AB . $\square$

7.2.1 Midpoints and Bisectors

The *midpoint* of a segment AB is the point M such that $AM \cong MB$ and $[AMB]$.

Axiom 11 (Midpoint Existence). $\forall A, B \in \mathbf{P}, A \neq B \implies \exists M \in \mathbf{P}: (\mathcal{B}(A, M, B)) \wedge (AM \cong MB)$.

Axiom 12 (Midpoint Uniqueness). $\forall A, B \in \mathbf{P}, A \neq B \implies \forall M, N \in \mathbf{P}: (AM \cong MB) \wedge (\mathcal{B}(A, M, B)) \wedge (AN \cong NB) \wedge (\mathcal{B}(A, N, B)) \implies M = N$.

<https://chat.deepseek.com/a/chat/s/c1efa70e-bf74-4e92-b967-4054f3472f26>

By removing betweenness as a criteria, we obtain the set of points $\{X \mid AM \cong MB\}$. This set is a line that is perpendicular to the segment AB . Alternatively, we can define the perpendicular bisector as the line containing the midpoint M of segment AB

7.3 Parallelism

Two lines are parallel if they have no point in common. In our set-theoretic foundations, this is to mean that the intersection of the lines is the empty set. Because we define parallelism as a binary relation on lines, we first must have a set of all lines to work with. Thus, we let $\mathbf{P}$ be the domain called the set of all points and define $\mathbf{L}$ to be the collection of all subsets of $\mathbf{P}$ that consist of the line $\overleftrightarrow{AB}$ for a any given two points $A, B \in \mathbf{P}$. Formally, that is:

$$\mathbf{L} := \{l \subseteq \mathbf{P} \mid \exists A, B \in \mathbf{P} (A \neq B \wedge l = \overleftrightarrow{AB})\}.$$

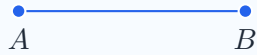
Definition 6 (Parallel). *Given any lines $l, m \in \mathbf{L}$, we say l is parallel to m and write $l \parallel m$ if and only if they are the same set of points or their set intersection is empty.*

$$\forall l, m \in \mathbf{L}, l \parallel m \iff (l = m) \vee (l \cap m = \emptyset).$$

add more here

Segment AB

$\{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}$



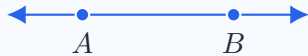
Ray $\overrightarrow{AB}$

Union of AB and points X where $\mathcal{B}(A, B, X)$



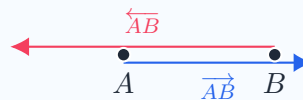
Line $\overleftrightarrow{AB}$

Extends infinitely in both directions



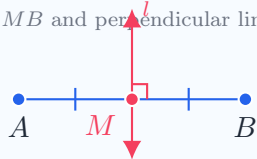
Intersection of Opposite Rays

$\overleftarrow{AB} \cap \overrightarrow{AB} = AB$



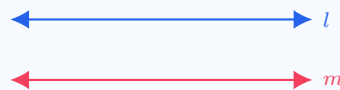
Midpoint & Bisector

$AM \cong MB$ and perpendicular line $l \perp AB$



Parallelism ($l \parallel m$)

No points in common: $l \cap m = \emptyset$



Chapter 8

Partition of the Plane

8.1 Plane Separation Axiom

The previous axioms of betweenness govern the linear ordering of points and are therefore limited to a single dimension. To extend this to two dimensions, we introduce the Plane Separation Axiom. This asserts that a line acts as a partition splitting the plane into two disjoint, non-overlapping regions, which is then used to describe a notion of points membership to one side of a given line through its logical equivalence to the negation of the previously defined notion of segment intersection.

Axiom 13 (Plane Separation). *For every line $l \in \mathbf{L}$, there exist two sets $H_{l,A}, H_{l,B} \subset \mathbf{P} \setminus l$ such that:*

1. $H_{l,A} \cup H_{l,B} = \mathbf{P} \setminus l \wedge H_{l,A} \cap H_{l,B} = \emptyset$.
2. $\forall A, B \in \mathbf{P} \setminus l: (\exists X \in l, \mathcal{B}(A, X, B)) \iff \{A, B\} \not\subset H_{l1} \wedge \{A, B\} \not\subset H_{l2}$.

explain the meaning-of-axioms-here

8.2 Same-Side Relation

We formalize the concept of 'being on the same side' through the Same-Side Relation ($\mathcal{S}$). It is defined by the absence of a separation/partition of the plane: two points are related if and only if unique segment connecting them does not contain any point of the boundary line l . This definition ensures the relation is based upon the primitive notion of betweenness. Specifically, we define $\mathcal{S}$ as a subset of the cartesian product $\mathcal{S} \subseteq \mathbf{P} \times \mathbf{P} \times \mathbf{L}$.

Definition 7 (Same-Side Relation). *Given a line $l \in \mathbf{L}$, any two points $A, B \in \mathbf{P} \setminus l$ are related by $\mathcal{S}$ if there does not exist a point $X \in l$ such that $\mathcal{B}(A, X, B)$. That is,*

$$\mathcal{S}(A, B, l) \iff \nexists X \in l: \mathcal{B}(A, X, B).$$

8.3 Half-Planes

By grouping all points related to a reference point A by $\mathcal{S}$, we construct the geometrical object known as the half-plane ($H_{l,A}$). This transition allows us to treat regions of the plane as sets that can be intersected to form more complex figures, such as angles and triangles.

Definition 8 (Half-Plane). *Let $l \in \mathbf{L}$ be a line and let $A \in \mathbf{P}$ be a reference point such that $A \notin l$. The half-plane determined by l and A , denoted $H_{l,A}$, is the subset of the point-set consisting of all points on the same side of l as A . That is,*

$$H_{l,A} := \{X \in \mathbf{P} \setminus l \mid \mathcal{S}(A, X, l)\}.$$

This geometric object allows us to move beyond comparisons between single points. A point X being on the same side as point Y is the same as X being an element of the half-plane $H_{l,y}$. This will show incredibly useful in defining derived geometric figures.

8.4 Theorems

Theorem 22 (Same-Side Reflexivity). *For any point $A \in \mathbf{P}$ that is not a member of any given line $l \in \mathbf{L}$, the following must hold:*

$$\forall l \in \mathbf{L}, \forall A \in \mathbf{P}: \mathcal{S}(A, A, l).$$

Proof. Let there be a line $l \in \mathbf{L}$ and a point $A \in \mathbf{P} \setminus l$. We proceed by contradiction. Assuming the same-side relation $\mathcal{S}(A, A, l)$ is false, it follows that $\nexists X \in l: \mathcal{B}(A, X, A)$ is false by definition, and that therefore there does exist an $X \in l$ satisfying these conditions. As a consequence, by the Identity/Reflexivity of Betweenness $A = X$.

However, we have defined that $A \notin l$. Since our assumption leads to a contradiction, it must be that no such point X exists. Thus, $\mathcal{S}(A, A, l)$ is true. $\square$

Theorem 23 (Same-Side Symmetry).

$$\mathcal{S}(A, B, l) \iff \mathcal{S}(B, A, l).$$

Proof. Firstly, we note that $\mathcal{S}(A, B, l) \iff \nexists X \in l: \mathcal{B}(A, X, B)$ by definition. By the Symmetry of Betweenness, the right-hand side of this is equivalent to $\nexists X \in l: \mathcal{B}(B, X, A)$, thus showing

$$X \in l: \neg \mathcal{B}(A, X, B) \iff X \in l: \neg \mathcal{B}(B, X, A).$$

The hand-side is by definition equivalent to $\mathcal{S}(B, A, l)$. Thus, by the transitivity of the biconditional, we conclude $\mathcal{S}(A, B, l) \iff \mathcal{S}(B, A, l)$. $\square$

Theorem 24 (Same-Side Transitivity). *Given any line $l \in \mathbf{L}$ and points $A, B, C \in \mathbf{P} \setminus l$, the relation $\mathcal{S}$ satisfies:*

$$\mathcal{S}(A, B, l) \wedge \mathcal{S}(B, C, l) \implies \mathcal{S}(A, C, l).$$

Proof. By the Plane Separation Axiom, we know there must exist two distinct sets H_{l1} and H_{l2} whose union is $\mathbf{P} \setminus l$. Since A, B , and C are not on l , each must belong to exactly one of these two sets.

Assume $\mathcal{S}(A, B, l)$. By definition, they are contained within the same set. Without loss of generality, let $A, B \in H_{l1}$.

Assume $\mathcal{S}(B, C, l)$. Similarly, this implies they are in the same set. We previously established that $B \in H_{l1}$, implying that $C \in H_{l1}$ by the plane separation axiom ($H_{l1} \cap H_{l2} = \emptyset$).

We now have $A \in H_{l1}$ and $C \in H_{l1}$. By the inverse of PSA2, there does not exist a point $X \in l$ such that $\mathcal{B}(A, X, C)$. Thus, by definition, A and C must be related by the same-side relation. That is, $\mathcal{S}(A, C, l)$. $\square$

randomideas:

Chapter 9

Angles

As opposed to being a measure, we define an angle as a geometric figure, ie: a set of points. Specifically, an angle is the union of two rays that share a common vertex.

Definition 9 (Angle). An *angle* is the union of two rays $\overrightarrow{AO}$ and $\overrightarrow{OB}$ that share a common vertex O , denoted as follows

$$\angle AOB := \overrightarrow{AO} \cup \overrightarrow{OB}.$$

Using the half-planes defined previously, we have the tools necessary to describe the interior and exterior of an angle.

Definition 10 (Interior of an Angle). For any three non-collinear points $A, B, C \in \mathbf{P}$, the *interior* of $\angle ABC$ is the set:

$$\text{Int}(\angle ABC) := \{P \in \mathbf{P} \mid \mathcal{S}(P, A, \overrightarrow{BC}) \wedge \mathcal{S}(P, C, \overrightarrow{BA})\}.$$

Equivalently, this is the intersection of two half-planes:

$$\text{Int}(\angle ABC) := H_{\overrightarrow{BC}, A} \cap H_{\overrightarrow{BA}, C}.$$

Definition 11 (Interior of an Angle). For any three non-collinear points $A, B, C \in \mathbf{P}$, the *interior* of $\angle ABC$ is the set:

$$\text{Int}(\angle ABC) := \{P \in \mathbf{P} \mid \mathcal{S}(P, A, \overrightarrow{BC}) \wedge \mathcal{S}(P, C, \overrightarrow{BA})\}.$$

Equivalently, this is the intersection of two half-planes:

$$\text{Int}(\angle ABC) := H_{\overrightarrow{BC}, A} \cap H_{\overrightarrow{BA}, C}.$$

Chapter 10

Triangles

10.1 Defintion

define as the union of three line segments. the three segments must be unique.

define inside and outside of the triangle based on if a point x is on the same side of each comprising segment or not

Chapter 11

Quadrilaterals

motivation

11.1 Definition

A quadrilateral is defined as the union of four unique noncollinear segments as follows:

$$\square ABCD := AB \cup BC \cup CD \cup DA.$$

Each segment is a side of the quadrilateral, and each point A, B, C, D is a vertex. Sides that share a vertex, such as AB and BC are adjacent, otherwise they are called opposite, such as AB and CD .

Theorem 25. *Each side of $\square ABCD$ contains its two endpoints and infinitely many interior points (by density).*

Theorem 26 (Adjacent sides intersect only at the common vertex). $AB \cap BC = \{B\}$, and similarly for the other three adjacent pairs.

Proof. Suppose a point $X \neq B$ belonged to both AB and BC . By the definitions of segment and betweenness, this forces A, B, C to be collinear, contradicting the non-collinearity condition. $\square$

11.2 Diagonals

Definition 12. *Given a quadrilateral $\square ABCD$, the segments AC and BD are the **diagonals** of $\square ABCD$.*

11.3 Trapezoid/Trapezium

We define a trapezoid as a quadrilateral with at least one pair of parallel sides. (In some traditions, a trapezoid has exactly one pair; we choose the inclusive definition, which places parallelograms as a special case of trapezoids.)

Definition 13. *A convex quadrilateral $\square ABCD$ is a trapezoid if and only if*

$$\overrightarrow{AB} \parallel \overrightarrow{CD} \text{ or } \overrightarrow{BC} \parallel \overrightarrow{DA}.$$

11.3.1 Diagonals**11.3.2 Isosceles Trapezoid**

A special case of the trapezoid occurs when the non-parallel sides (legs) are congruent. If this is the case, the quadrilateral is an isosceles trapezoid.

Definition 14 (Isosceles Trapezoid). *A trapezoid $\square ABCD$ with $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$ a isosceles trapezoid if and only if*

$$AD \cong BC.$$

Chapter 12

Quadrilaterals

motivation

12.1 Definition

A quadrilateral is defined as the union of four unique noncollinear segments as follows:

$$\square ABCD := AB \cup BC \cup CD \cup DA.$$

Each segment is a side of the quadrilateral, and each point A, B, C, D is a vertex. Sides that share a vertex, such as AB and BC are adjacent, otherwise they are called opposite, such as AB and CD .

Theorem 27. *Each side of $\square ABCD$ contains its two endpoints and infinitely many interior points (by density).*

Theorem 28 (Adjacent sides intersect only at the common vertex). $AB \cap BC = \{B\}$, and similarly for the other three adjacent pairs.

Proof. Suppose a point $X \neq B$ belonged to both AB and BC . By the definitions of segment and betweenness, this forces A, B, C to be collinear, contradicting the non-collinearity condition. $\square$

12.2 Diagonals

Definition 15. *Given a quadrilateral $\square ABCD$, the segments AC and BD are the **diagonals** of $\square ABCD$.*

12.3 Trapezoid/Trapezium

We define a trapezoid as a quadrilateral with at least one pair of parallel sides. (In some traditions, a trapezoid has exactly one pair; we choose the inclusive definition, which places parallelograms as a special case of trapezoids.)

Definition 16. *A convex quadrilateral $\square ABCD$ is a trapezoid if and only if*

$$\overrightarrow{AB} \parallel \overrightarrow{CD} \text{ or } \overrightarrow{BC} \parallel \overrightarrow{DA}.$$

12.3.1 Diagonal

12.3.2 Isosceles Trapezoid

A special case of the trapezoid occurs when the non-parallel sides (legs) are congruent. If this is the case, the quadrilateral is an isosceles trapezoid.

Definition 17 (Isosceles Trapezoid). *A trapezoid $\square ABCD$ with $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$ a isosceles trapezoid if and only if*

$$AD \cong BC.$$

12.3.3 Right Trapezoid

Definition 18 (Right trapezoid). *A quadrilateral $ABCD$ is a right trapezoid if and only if it is a trapezoid and at least one of its interior angles is a right angle (if and only if at least one of its legs is perpendicular to the bases.)*

12.4 parallelograms

Theorem 29. *The diagonals of an isosceles trapezoid are congruent.*

Theorem 30 (Diagonals of isosceles trapezoid are congruent). *Given an isosceles trapezoid $\square ABCD$ with $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$, the diagonals are congruent:*

$$AC \cong BD.$$

12.5 Special Parallelograms

12.5.1 Rectangles

12.5.2 Rhombus

Definition 19 (Rhombus is a parallelogram whose sides are all the same length). *A rhombus is a parallelogram whose four sides are congruent to one another.*

Definition 20 (Rhombus is Equilateral but not Equiangular). *me e9tuer Rhombus is Equilateral but not Equiangular*

12.5.3 Square

12.6 Parallelograms

Parallelogram A quadrilateral is a parallelogram if both pairs of opposite sides are parallel.

Definition 21 (Parallelogram). *Let $A, B, C, D \in \mathbf{P}$ be four points satisfying the quadrilateral conditions. Then the quadrilateral $\square ABCD$ is a **parallelogram** if and only if*

$$\overleftrightarrow{AB} \parallel \overleftrightarrow{CD} \text{ and } \overleftrightarrow{BC} \parallel \overleftrightarrow{DA}.$$

i hate this project lmao ts straight retarded. idk what to do with it ever.

12.7 Special Parallelograms

12.7.1 Rectangles

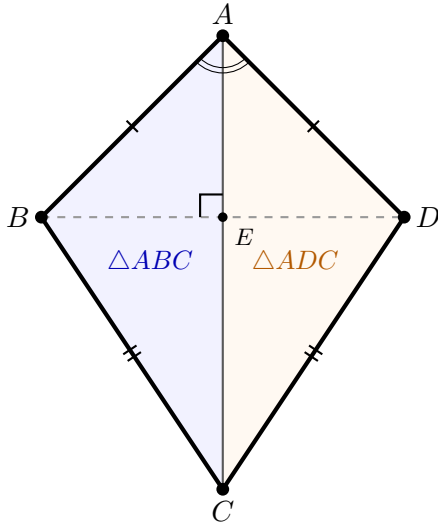
12.7.2 Rhombus

12.7.3 Square

12.8 Kites

Definition 22 (Kite). A convex quadrilateral $\square ABCD$ is a *kite* if

$$AB \cong AD \wedge CB \cong CD.$$



Theorem 31. In a kite $\square ABCD$,

$$\angle BAC \cong \angle DAC \text{ and } \angle BCA \cong \angle DCA.$$

Proof. Consider the triangles $\triangle ABC$ and $\triangle ADC$.

- $AB \cong AD$ by definition
- $CB \cong CD$ by definition
- $AC \cong AC$ by reflexivity of congruence

By the SSS triangle congruency criteria, $\triangle ABC \cong \triangle ADC$. Consequently, the corresponding angles are congruent:

$$\angle BAC \cong \angle DAC \text{ and } \angle BCA \cong \angle DCA.$$

□

Theorem 32 (The axis diagonal is the perpendicular bisector of the cross diagonal). In a kite $\square ABCD$, let $E = AC \cap BD$. Then,

$$AC \perp BD \text{ and } BE \cong ED.$$

Proof. wasd □

Axiom 14 (Segment Extension with Betweenness). *For any A, B, C, D there exists E such that $H(A, B, C, E, D)$.*

Axiom 15 (Uniqueness of Extension). *If $H(A, B, C, E, D)$ and $H(A, B, C, E', D)$, then $E = E'$.*

Axiom 16 (Symmetry of Congruence). $H(A, B, C, D, E) \implies H(C, D, A, E, B)$

Theorem 33 (Midpoint quadrilateral of a kite). *The quadrilateral formed by joining the midpoints of the sides of a kite is a rectangle.*

Proof. □

12.8.1 Right Kite

12.9 Midpoint Theorems

12.10 Cyclic Quadrilaterals

12.11 Tangential Quadrilaterals

12.12 Bicentric Quadrilaterals

12.13 Other

11.15 Other Niche Quadrilaterals (Optional / Encyclopaedic) 11.15.1 Equidiagonal quadrilaterals (diagonals equal in length).

11.15.2 Orthodiagonal quadrilaterals (diagonals perpendicular).

11.15.3 Harmonic quadrilaterals (vertices form a harmonic division on the circumcircle – requires cross-ratio, so likely beyond your current primitives, but can be mentioned).

11.15.4 Quadrilaterals with special area properties (e.g., Brahmagupta's formula for cyclic quadrilaterals).

11.15.5 Complete quadrilaterals (the figure formed by four lines, six intersection points – a projective concept; may be out of scope).

11.15.6 Miquel point of a complete quadrilateral (four circles concurrent).

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